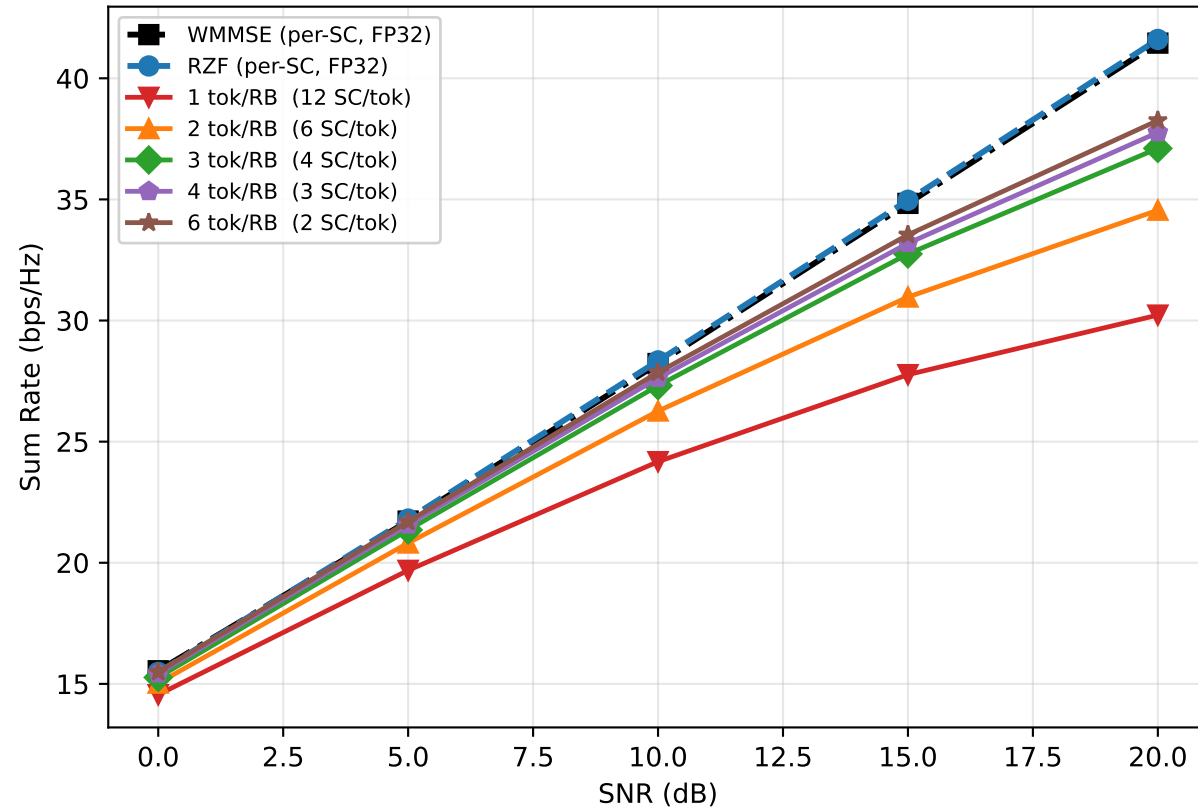


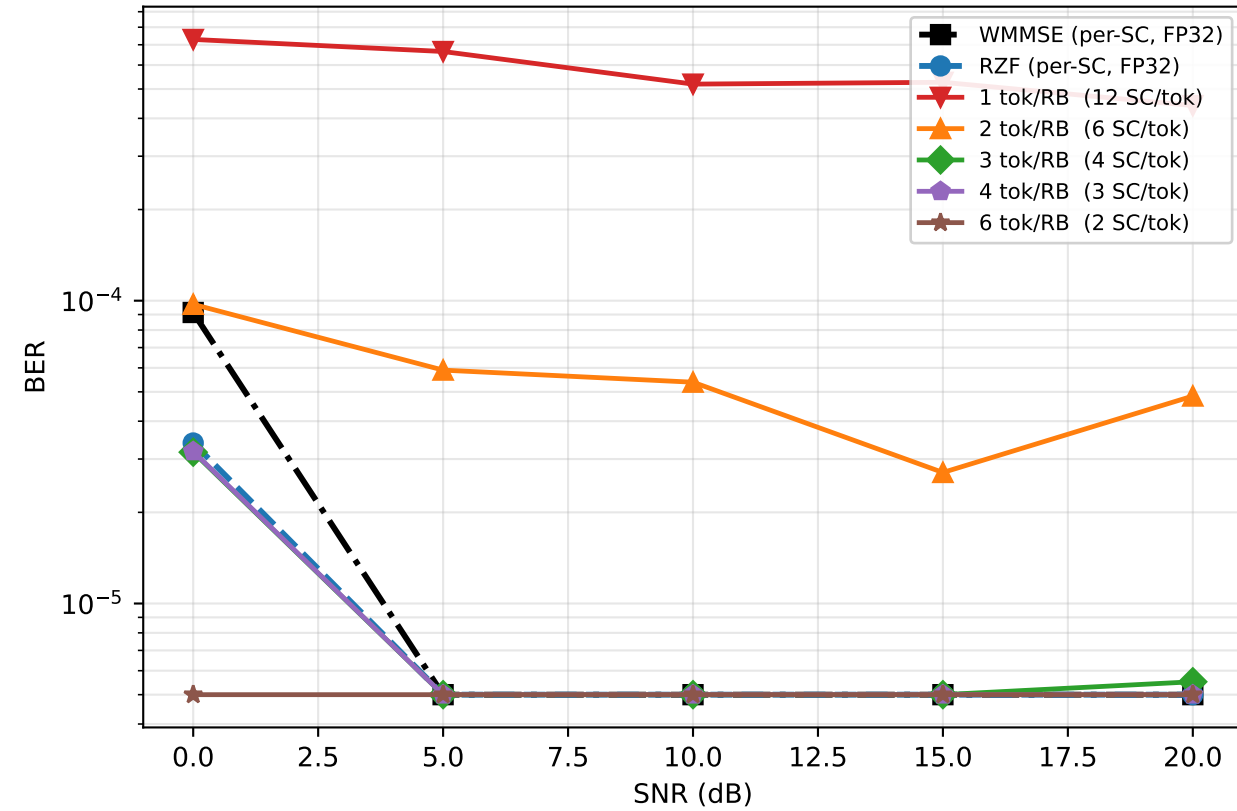
V4.2 Transformer Precoder — 8×4 MU-MIMO

Ablation: tokens per Resource Block (RB = 12 SC)

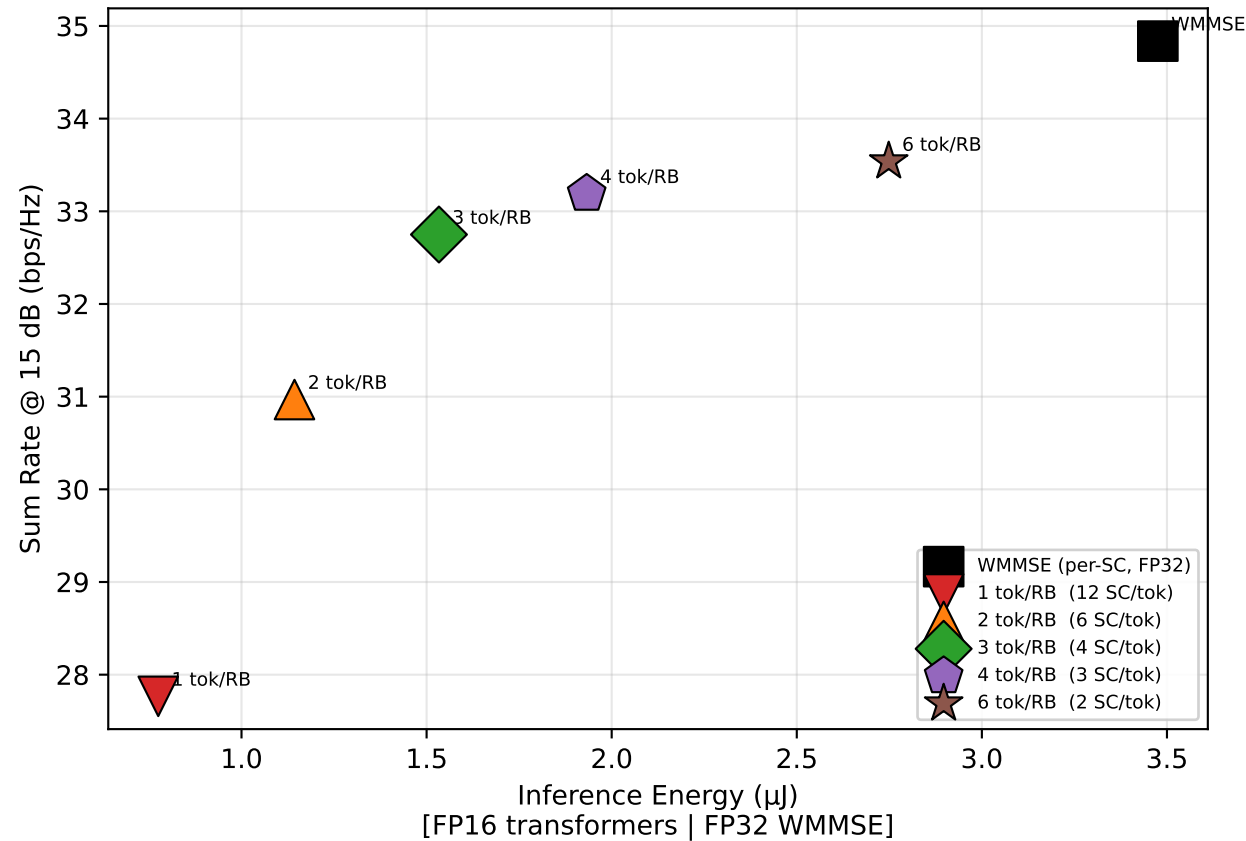
(a) Spectral Efficiency



(b) Bit Error Rate (Monte Carlo)



(c) Energy-Performance Pareto



(d) Summary @ 15 dB
Gain = WMMSE_FP32 / model_FP16

Method	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	FP16 (μ J)	Gain vs WMMSE
RZF (per-SC, FP32)	34.97	+0.13	2	N/A	ref
WMMSE (per-SC, FP32)	34.84	+0.00	1842	N/A	ref
6 tok/RB (2 SC/tok)	33.54	-1.30	5080	2.7484	1.3×
4 tok/RB (3 SC/tok)	33.19	-1.65	3571	1.9325	1.8×
3 tok/RB (4 SC/tok)	32.75	-2.09	2833	1.5334	2.3×
2 tok/RB (6 SC/tok)	30.97	-3.87	2112	1.1431	3.0×
1 tok/RB (12 SC/tok)	27.77	-7.07	1432	0.7751	4.5×